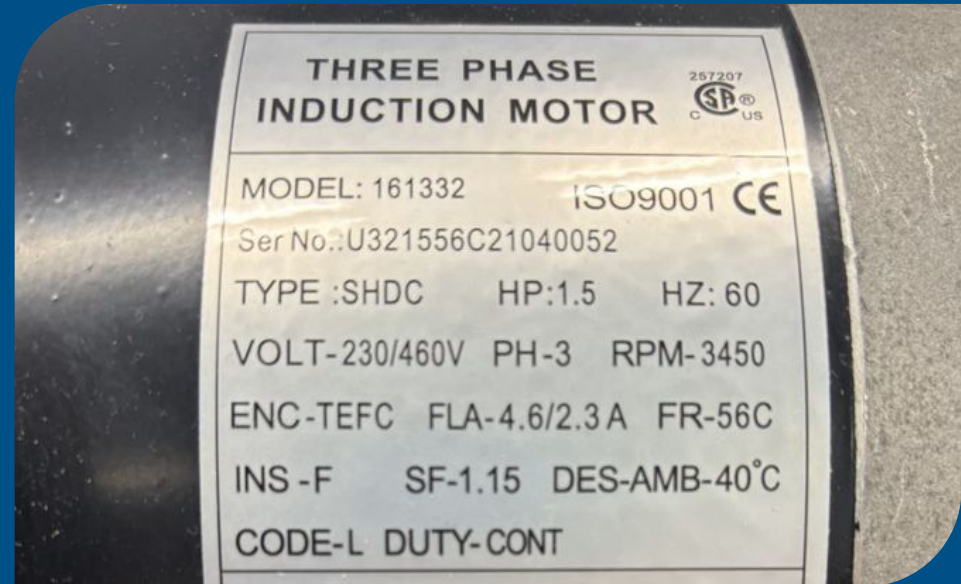


3-Phase AC Motor

By: Henry Matar & Neill Raymond



Topic Description

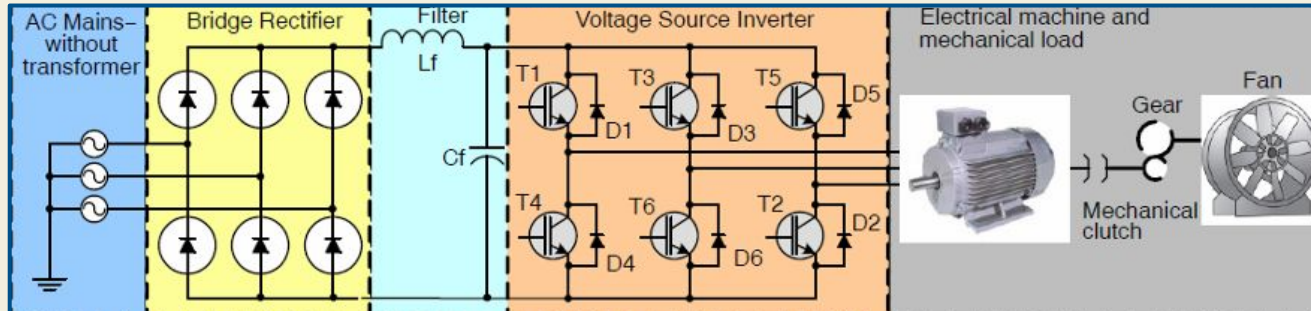


Given:

- 3-Phase, 230/460V, 1.5HP, 2-Pole Motor
- Variable Frequency Drive (VFD)
- 120V Single Phase Source

Goals:

- Measure Physical Parameters
- Solve for other Parameters
- Create a Model of Motor



Taking Measurements

Supplies:

- Scale
- Vice Grips
- Ratchet Strap
- Tachometer
- Multimeter & Clamp Meter

Rotor Shaft Length – 8.25in

Number of Poles – 2

Two Key States

- **Free Spinning:**
 - Supply Voltage
 - Supply Current
 - Rotational Speed
 - Line Voltage
 - Line Current
- **Stalled Rotor:**
 - Supply Current
 - Line Voltage
 - Down Force Applied

Measurement Results

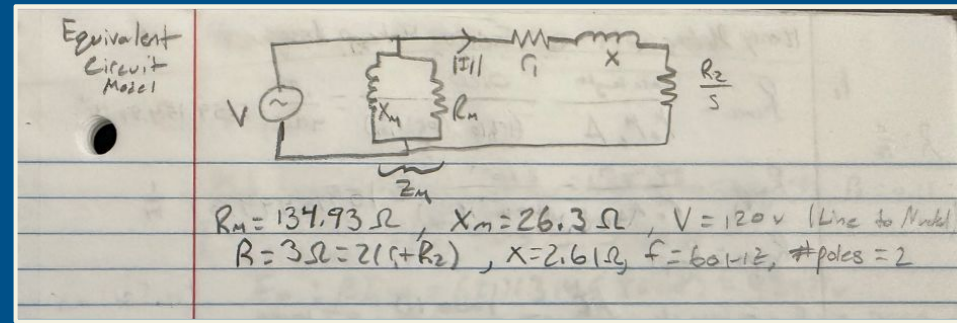
- Measure at different frequencies
- Motor RPM closely matches expected
- Large source current when stalled

3-PHASE VFD DRIVEN MOTOR CALCULATIONS				
Frequency - [Hz]	10	20	30	40
MEASUREMENT	MEASURED VALUES - FREE SPINNING			
Single Phase Voltage - [V]	120	120	120	120
Single Phase Current - [A]	0.9	1.2	1.6	1.9
Motor Speed - [RPM]	592	1196	1793	2387
L1-L2 Voltage - [V]	122.5	167.5	203.0	233.0
L1-L3 Voltage - [V]	122.5	167.5	203.0	233.0
L2-L3 Voltage - [V]	122.5	167.5	203.0	233.0
L1 Current - [A]	1.3	2.0	2.3	2.5
L2 Current - [A]	1.3	2.0	2.3	2.5
L3 Current - [A]	1.3	2.0	2.3	2.5
MEASUREMENT	MEASURED VALUES - STALLED ROTOR (I=8.25in)			
Single Phase Current - [A]	5.2	17.4	28.0	36.0
L1-L2 Voltage - [V]	105	133	140	143
L1-L3 Voltage - [V]	105	133	140	143
L2-L3 Voltage - [V]	105	133	140	143
Down Force - [g]	1570	2823	2917	2743

Making a Model

Matlab Script:

- Enter Motor Parameters
- Compute sync speeds & V_{L-L}
- Sweep from Slip 0 $\rightarrow$ 1
- Solve Rotor Branch
- Solve Magnetizing Branch
- Solve Stator Branch
- Calculate Torque
- Calculate Mechanical Power Out



```
%% Equivalent Circuit Calculations
for k = 1:length(s)
    sk = s(k);

    % Rotor branch
    Z2 = R2/sk + 1j*x;

    % Magnetizing branch
    Zm = 1j*Xm;

    % Stator branch
    Z1 = r + 1j*x;

    % Thevenin Voltage
    Vth(k) = V_phase * (Zm)/(Z1 + Zm);

    % Thevenin Impedance
    Zth(k) = (Z1 * Zm)/(Z1 + Zm);

    % Extract Rth and Xth
    Rth = real(Zth(k));
    Xth = imag(Zth(k));

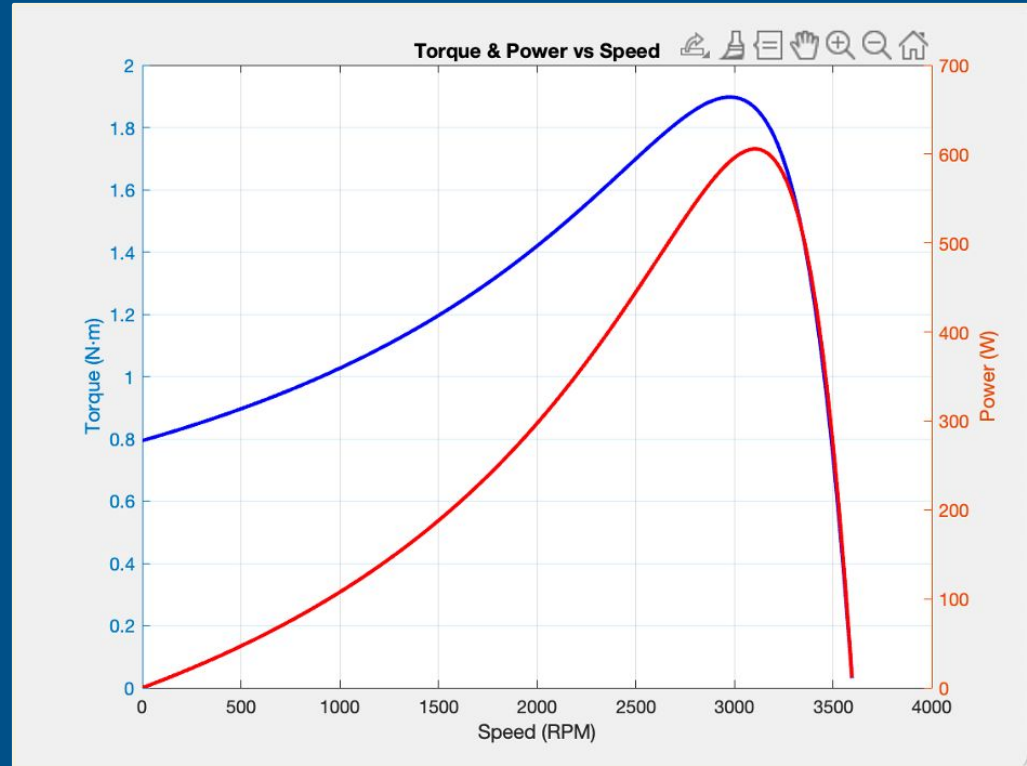
    % Torque expression with slip-dependent rotor resistance
    Tor(k) = 3 * abs(Vth(k))^2 * (R2/sk) / ...
    (wsync * ((Rth + R2/sk)^2 + (Xth + x)^2));
end

%% Mechanical speed
n_rpm = (1 - s) * nsync_rpm;
w_mech = n_rpm * (2*pi/60);

%% Output power
P_out = Tor .* w_mech; % W
```

Initial Model

- Using common values for unknowns
- Verifies model validity



Solving for Magnetization Branch Impedances for L and R_M

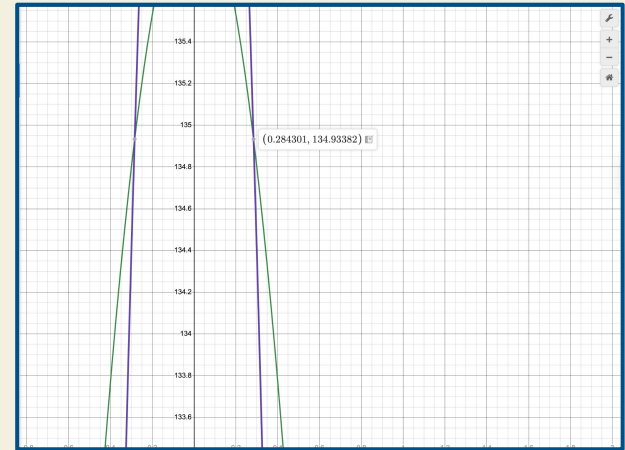
Frequency (Hz)	V (v)	I (A)	Z _m (Ω)
10	122.5	0.9	136.11
20	167.5	1.2	139.58
30	203	1.6	126.86
40	233	1.9	122.63

$$Z_m = R_m + jX_m$$

$$|Z_m| = \sqrt{R_m^2 + (2\pi fL)^2}$$

$$|V| = |I||Z_m|$$

$$\frac{|V|}{|I|} = \sqrt{R_m^2 + (2\pi fL)^2}$$



- L = 284.3mH
- R_M = 134.93Ω

Locked Rotor Torque and Current – Solving R_2

f (Hz)	T_{LR} (Nm)	ω_s (rad/sec)	R_2 (Ω)
10	3.227	62.83	7.498
20	5.803	125.66	2.409
30	5.996	188.50	1.44
40	5.639	251.33	1.09

$$T_{LR} = \frac{|I_1|^2}{\omega_s} R_2$$

$$I = \frac{V}{Z_m} + I_1$$

$$I_1 = \frac{V}{r_1 + R_2 + jx}$$

- Compare to Line to Line Resistance
 - $R = 2(r_1 + R_2) = 3$
- Magnetization current is small
- Selected Values of R_2 to solve for I_1
- Use R_2 for more accurate values for I_1
 - Don't need to approximate $I = I_1$
 - $I_1 = I - V/Z_m$

Solving for I_1 and X

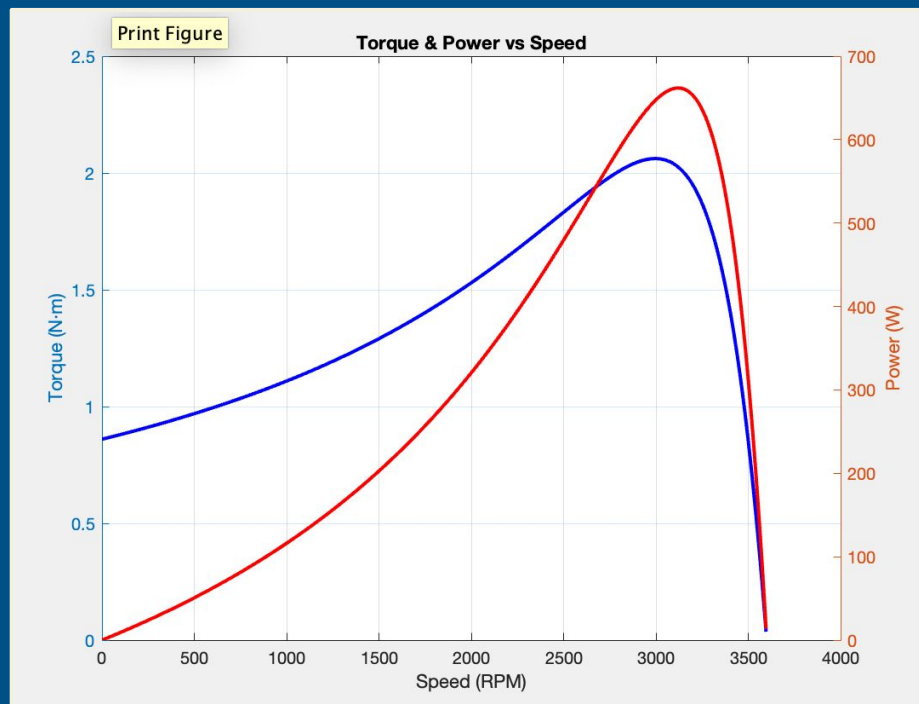
f (Hz)	I1 (A)
30	26.40
40	34.83

f (Hz)	x (Ω)
30	3.8
40	2.61

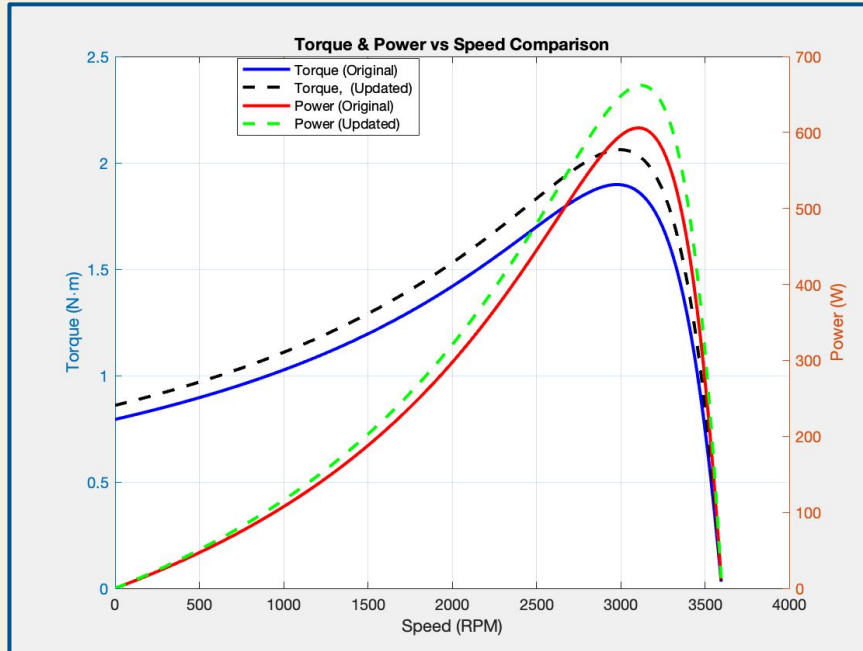
- $T_{LR} = \frac{|I_1|^2}{\omega_s} R_2$
- Find X:
- Every parameter now found for x
 - $I_1 = \frac{V}{r_1 + R_2 + jx}$

Refined Model

- New Torque and Power Curve with Solved Parameters
- Increase in Torque and Power compared to initial model
 - Closer to Listed Specs



Comparison of Measurements to Simulations



- Compares initial model with model derived from experimental measurements at 30, 40 Hz
- Power output approaches rated 1.1kW maximum.
- Yield power is less than maximum rating
 - Likely due to measurements under 60Hz and 230V source

Conclusion

From just the motor and its name plate, we got:

- Physical measurements
- Parametrized motor equivalent circuit
- Motor Model Script
- Power & Torque Plots

